Instructions:

The student must recreate the scenario through Packet tracer software version 8.1.1.0022 according to the study case denominated Lab-Ten and email the packet tracer files. Email: escajadillomunoa@rowan.edu with the subject: Lab-10

Study Case: Lab N° 10

Topic: Basic configuration of Firewall ASA 5505

Scenario-1:

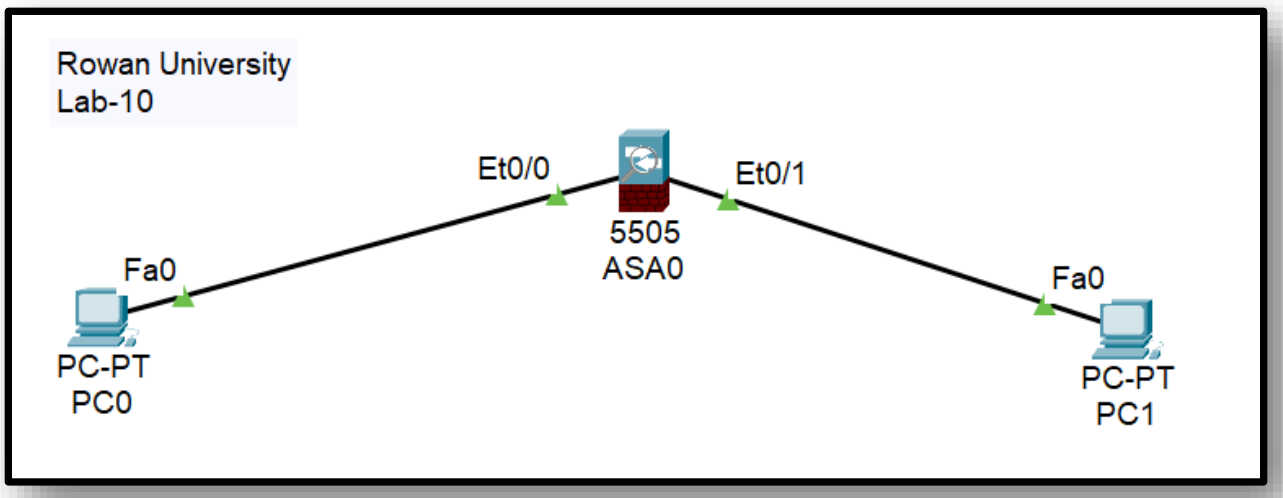
Creating a network design called L-10-1 that contains the following steps:

Step 1:Select components

Select the three components below and connect through Copper Straight-Through (See Fig N° 1).

- PC-PT PC0--->FastEthernet0 (Fa0)
- PC-PT PC1--->Fast Ethernet0 (Fa0)
- 5505 ASA0--->Ethernet0/0 (Et0/0) and ---> Ethernet0/1 (Et0/1)

Fig N° 1



Step 2:Assignent IP on PC0 and PC1

Click on PC-PT PC[0] and PC[1], Desktop-Tab, IP configuration, select the radio button Static, and set up the following configuration below:(See Fig N° 2A and 2B).

PC-PT PC0---> IP : 192.168.1.2
Subnet Mask : 255.255.255.0
Default Gateway : 192.168.1.1

PC-PT PC1---> IP : 192.168.2.2
Subnet Mask : 255.255.255.0
Default Gateway : 192.168.2.1

Fig N° 2A

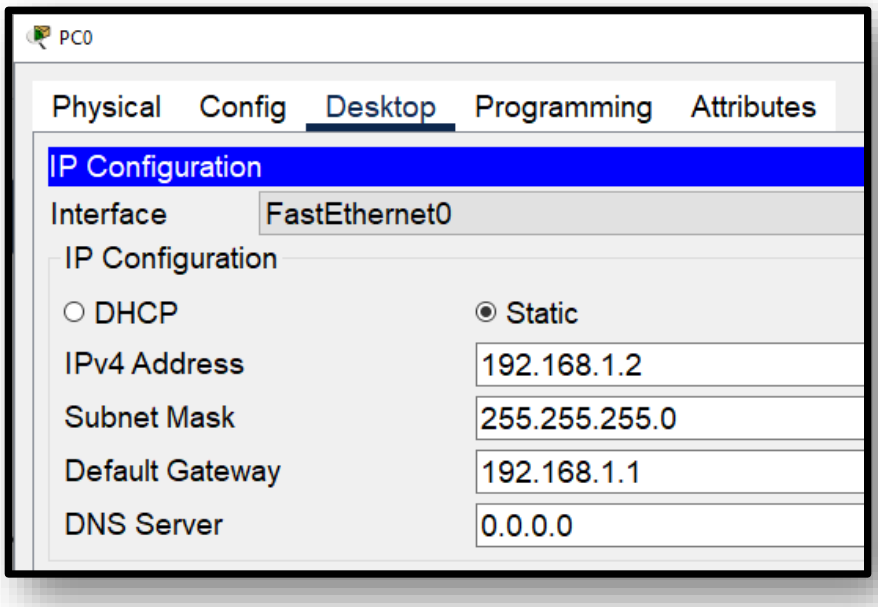
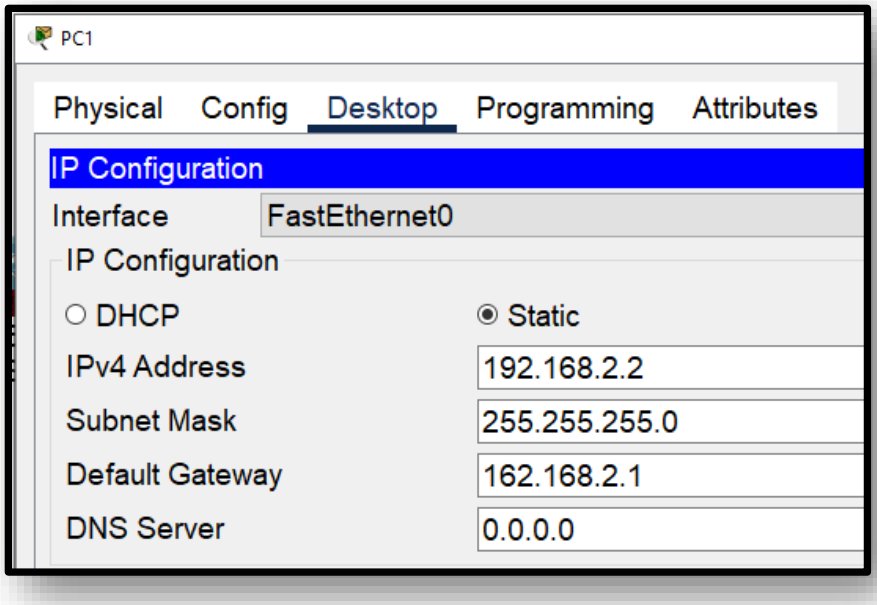


Fig N° 2B

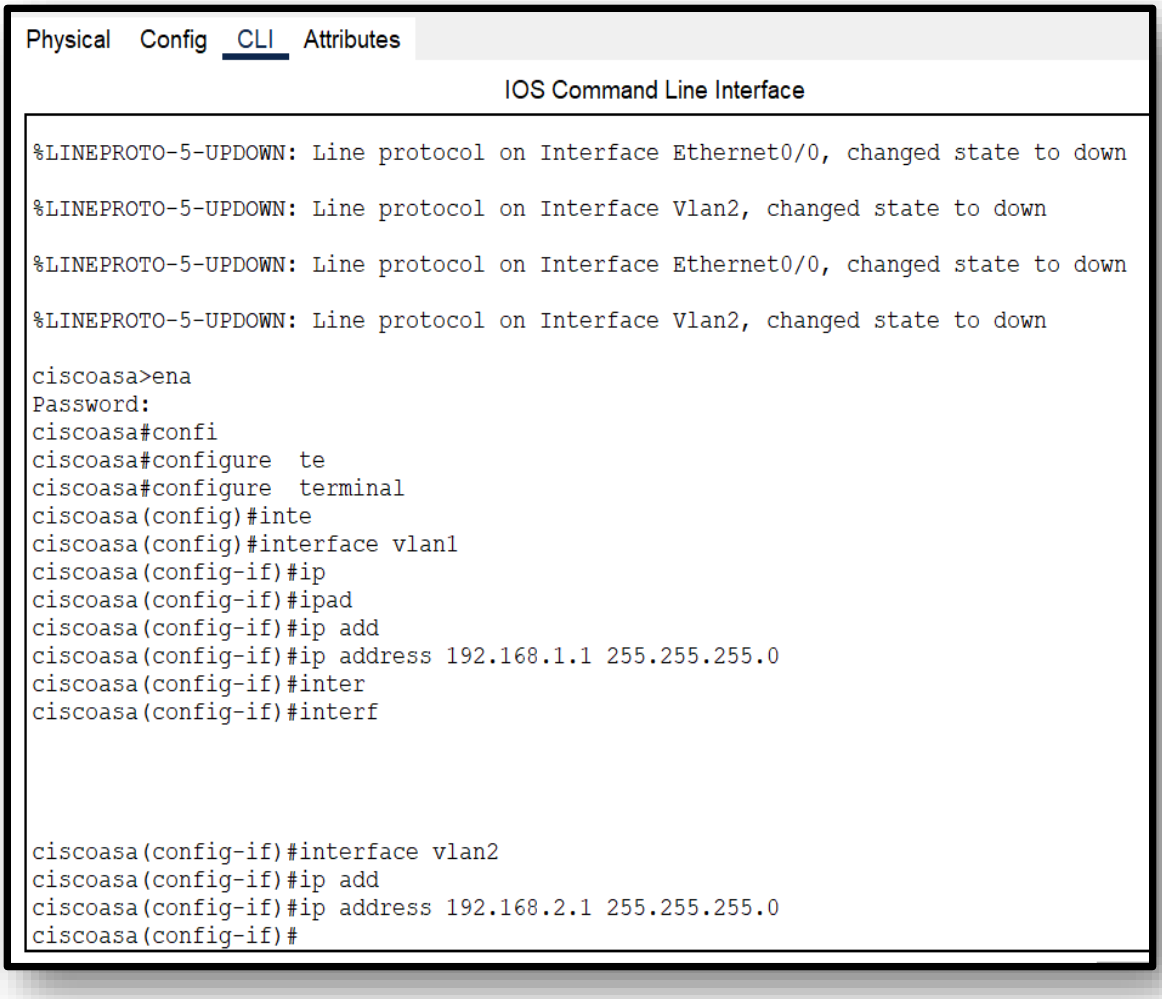


Step 2:Set up ASA 5505
click on 5505 ASA0, CLI-Tab, and set up the following configuration below

VLAN configuration :(See Fig N° 3A)

```
enabled
configure terminal
interface vlan1
ip address 192.168.1.1 255.255.255.0
interface vlan2
ip address 192.168.2.1 255.255.255.0
```

Fig N° 3A



Port configuration :(See Fig N° 3B)

```
interface ethernet 0/0
switchport access vlan 1
no shutdown
interface ethernet 0/1
switch access vlan 2
no shutdown
exit
exit
exit
```

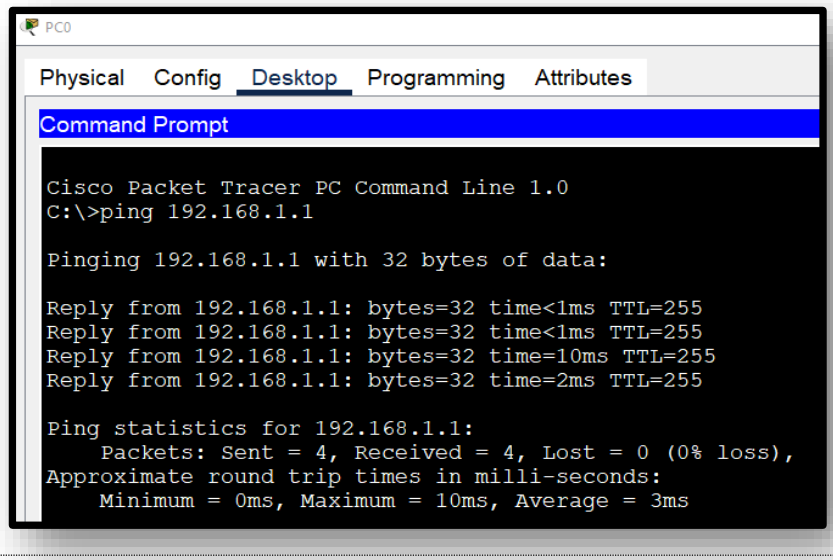
Fig N° 3B

```
ciscoasa(config-if)#interface ethernet 0/0
ciscoasa(config-if)#switchport access vlan 1
ciscoasa(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to down
no shutdown
ciscoasa(config-if)#interface ethernet 0/1
ciscoasa(config-if)#switchport access vlan 2
ciscoasa(config-if)#no shutdown
ciscoasa(config-if)#exit
ciscoasa(config)#exit
ciscoasa#exit
```

Step 3: Testing Set up ASA 5505
Click on PC0, Desktop-Tab, Command Prompt, and follow the testing below.

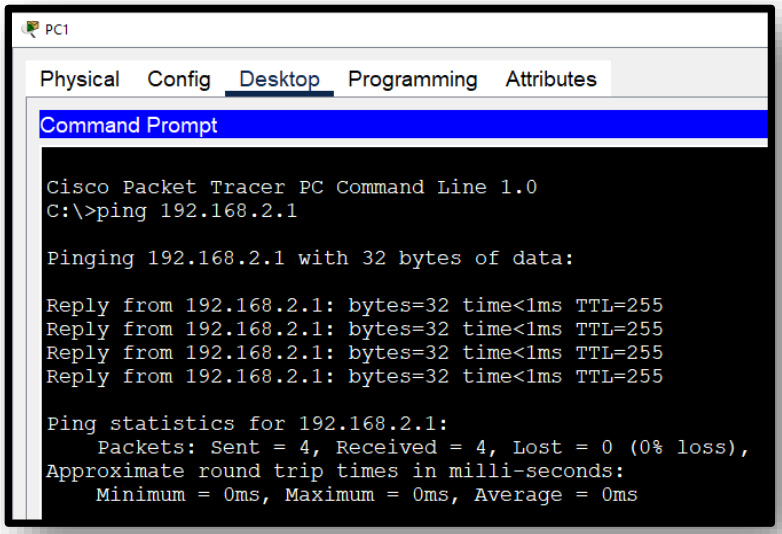
Testing from PC0 to ASA. (See Fig N° 4A)

Fig N° 4A



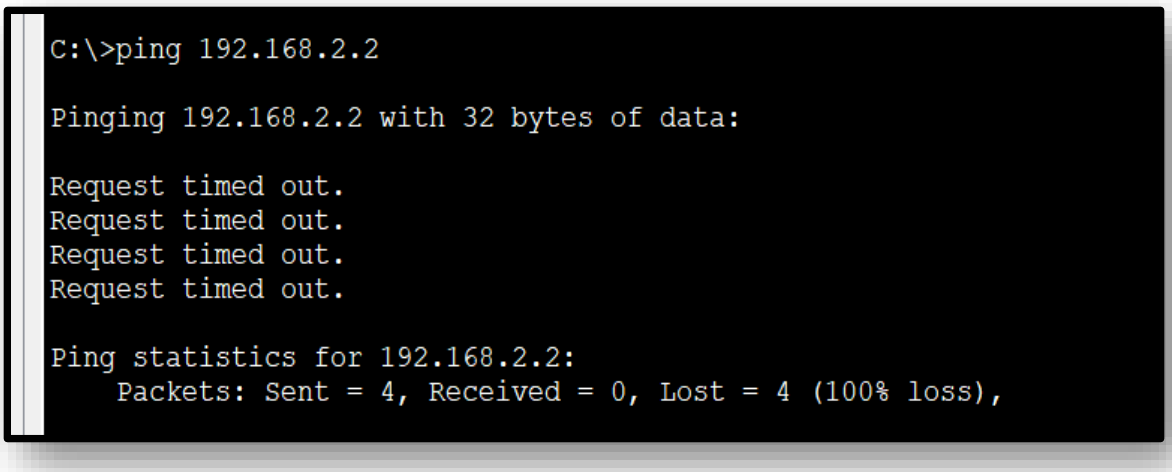
Testing from PC1 to ASA. (See Fig N° 4B)

Fig N° 4B



Testing from PC0 to PC1. (See Fig N° 4C)

Fig N° 4C



Step 4: Set up rules or policy at ASA 5505
Click on ASA 5505, Cli-Tab, and follow the configuration below

Configuration-Policy :(See Fig N° 5A,5B,5C,and 5D)

```
enabled
configure terminal
class-map UES1
match default-inspection-traffic
exit
policy-map policy1
class UES1
inspect icmp
exit
service-policy policy1 global
show running
```

Fig N° 5A

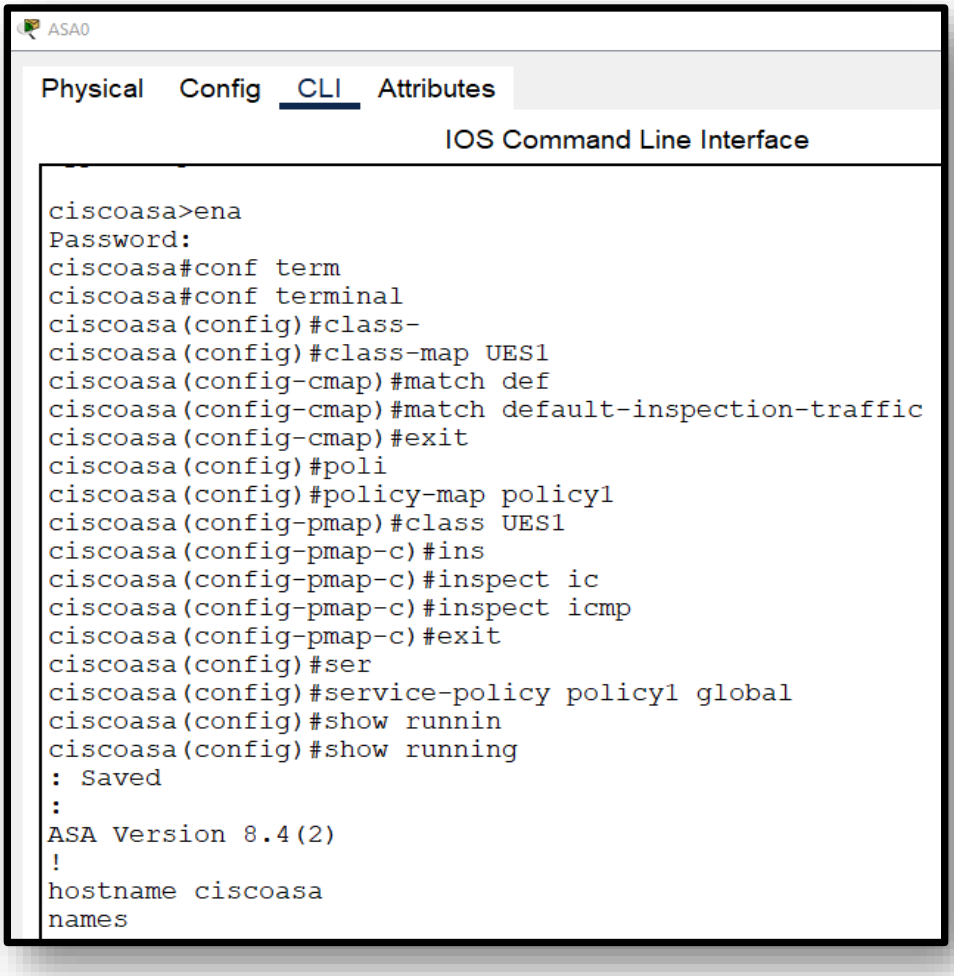


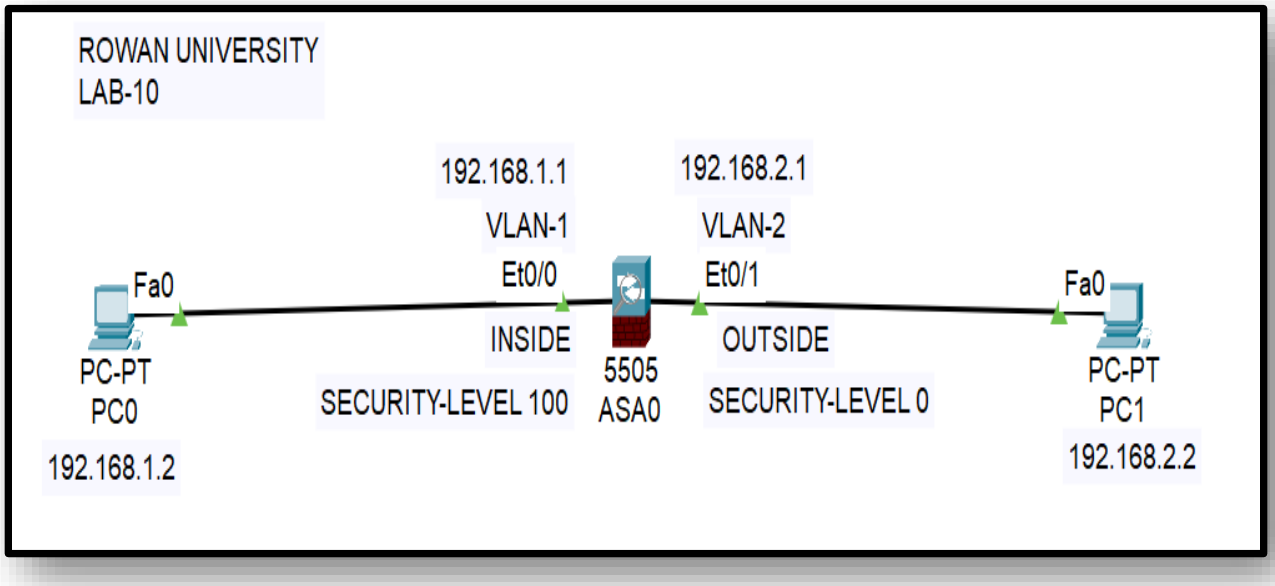
Fig N° 5B

```
!
interface Ethernet0/0
!
interface Ethernet0/1
 switchport access vlan 2
!
interface Ethernet0/2
!
interface Ethernet0/3
!
interface Ethernet0/4
!
interface Ethernet0/5
!
interface Ethernet0/6
!
interface Ethernet0/7
!
interface Vlan1
 nameif inside
 security-level 100
 ip address 192.168.1.1 255.255.255.0
!
interface Vlan2
 nameif outside
 security-level 0
 ip address 192.168.2.1 255.255.255.0
!
```

Fig N° 5C

```
!
class-map UES1
 match default-inspection-traffic
!
policy-map policy1
 class UES1
  inspect icmp
!
service-policy policy1 global
!
telnet timeout 5
ssh timeout 5
!
dhcpd auto_config outside
!
dhcpd address 192.168.1.5-192.168.1.36 inside
dhcpd enable inside
!
!
!
!
!
ciscoasa(config)#
```

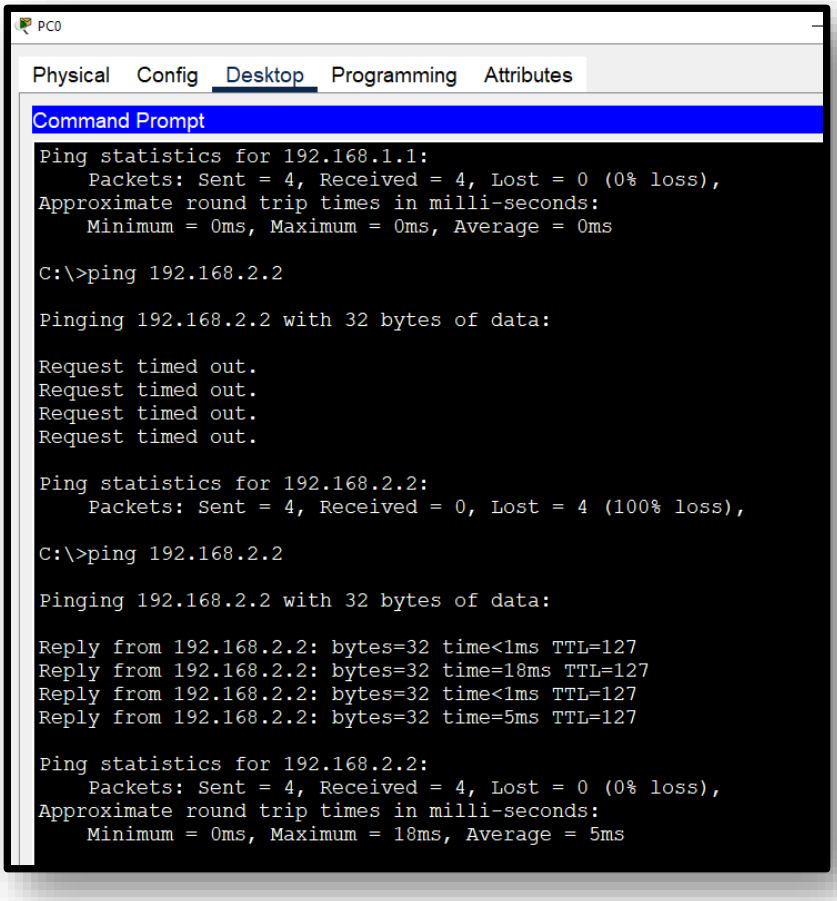
Fig N° 5D



Step 5: Testing Communication successfully from Inside to Outside

Click on PC0, Cli-Tab, and follow the configuration below.:
(See Fig N° 6)

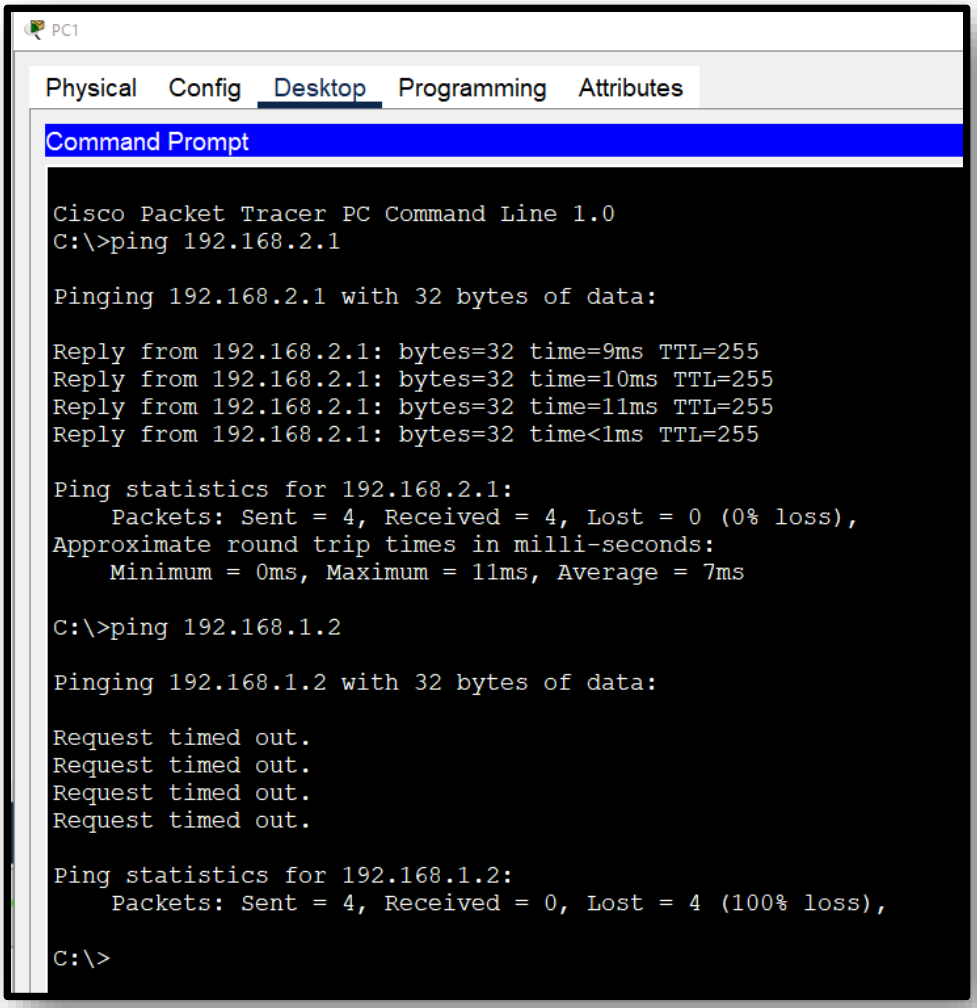
Fig N° 6



Step 5: Testing Communication not authorized from Outside to inside

Click on PC1, Cli-Tab, and follow the configuration below.:
(See Fig N° 7)

Fig N° 7



Step 5: Simulation from inside to outside and vice versa
(See Fig N° 8A and 8B)

Fig N° 8A

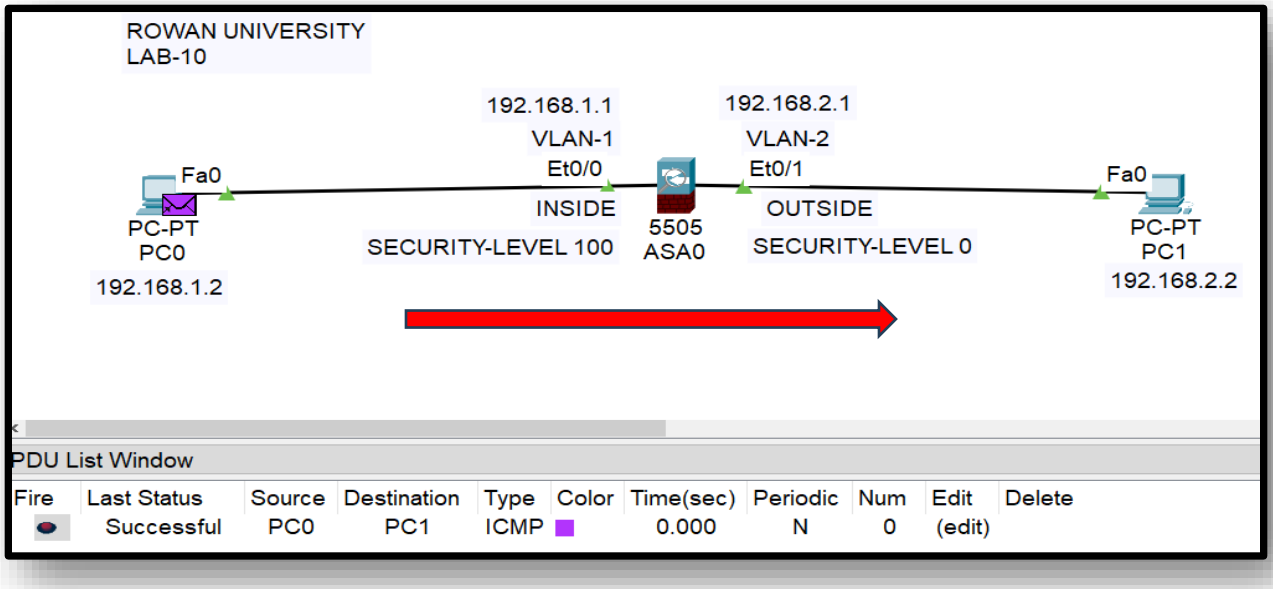
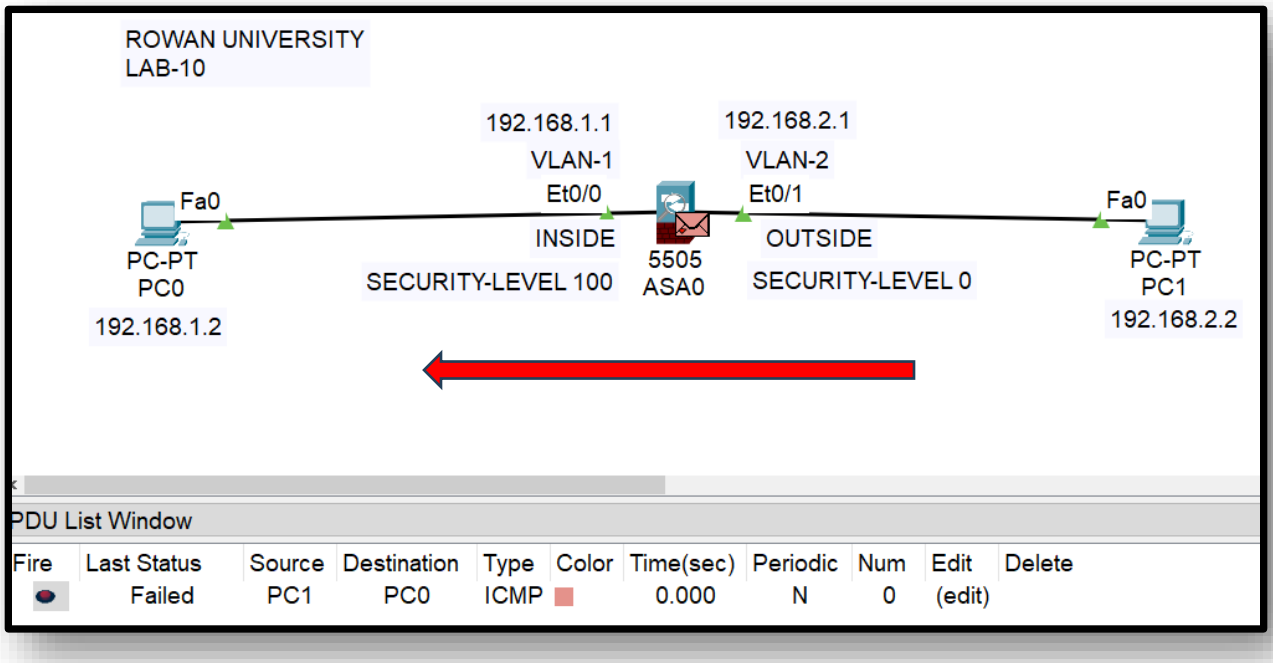


Fig N° 8B



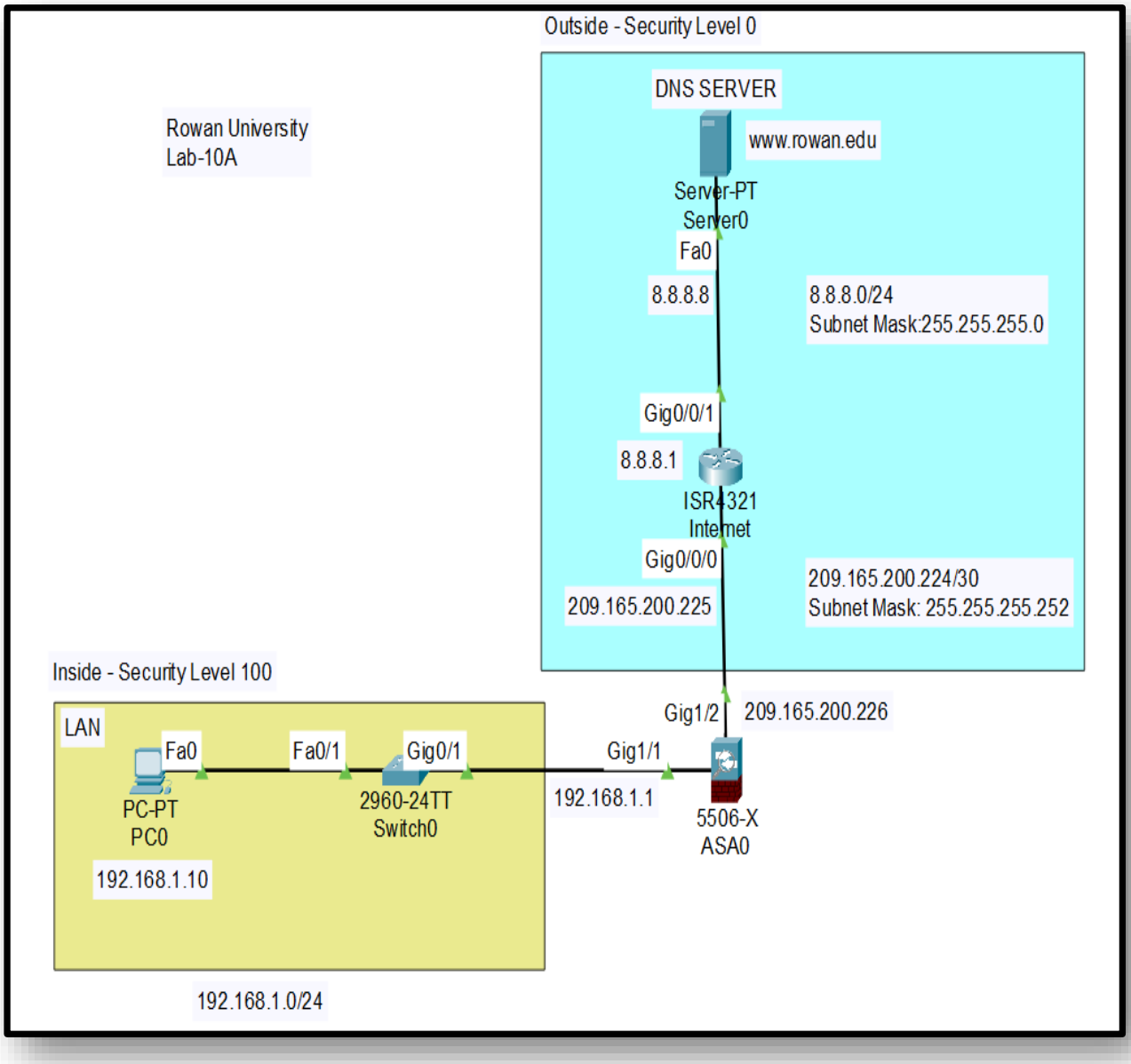
Lab N° 10A

Topic: configuration of Firewall ASA 5505 (DHCP, route, NAT, Inspect, SSH)

Scenario-2:

Creating a network design called L-10-2 that contains the following steps:

Fig. Scenario-2

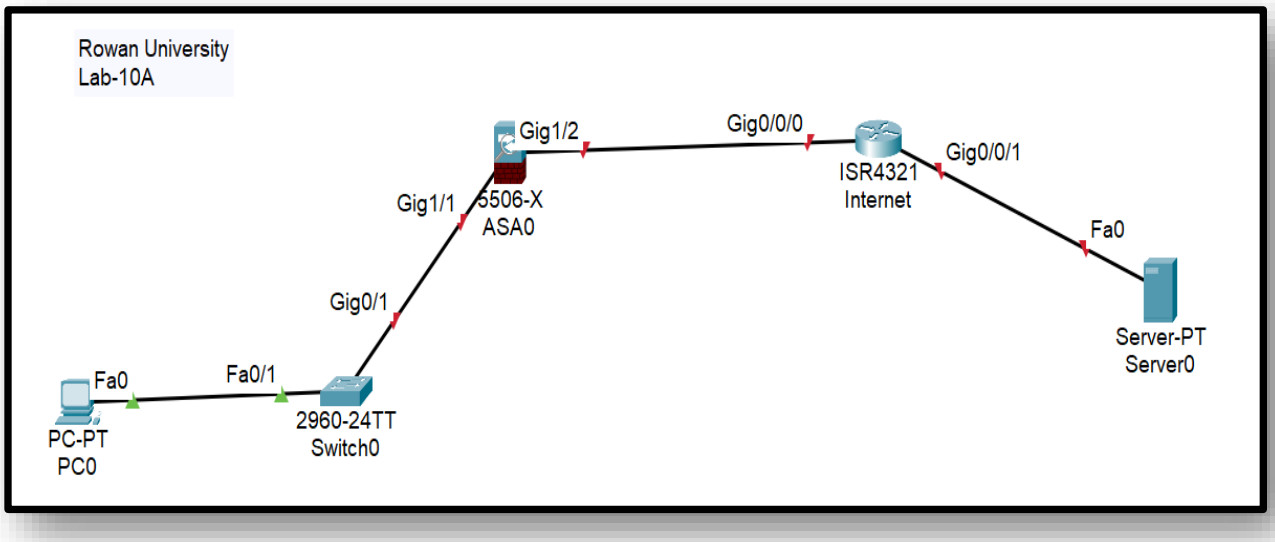


Step 1:Select components

Select the components below and connect through Copper Straight-Through (See Fig N° 1).

- PC-PT PC0---------->FastEthernet0 (Fa0)
- 2660-24TT SWITCH0------->Fast Ethernet0/1 (Fa0/1) and GigabitEthernet0/1 (Gib0/1)
- 5506-X ASA0---------->GigabitEthernet1/1 (Gib1/1)and GigabitEthernet1/2 (Gib1/2)
- (Router)ISR4321"Internet"------>GigabitEthernet0/0/0 (Gig0/0/0) and GigabitEthernet0/0/1 (Gib 0/0/1)
- Servet-PT Server 0---->Ethernet0 (Fa0)

Fig N° 1

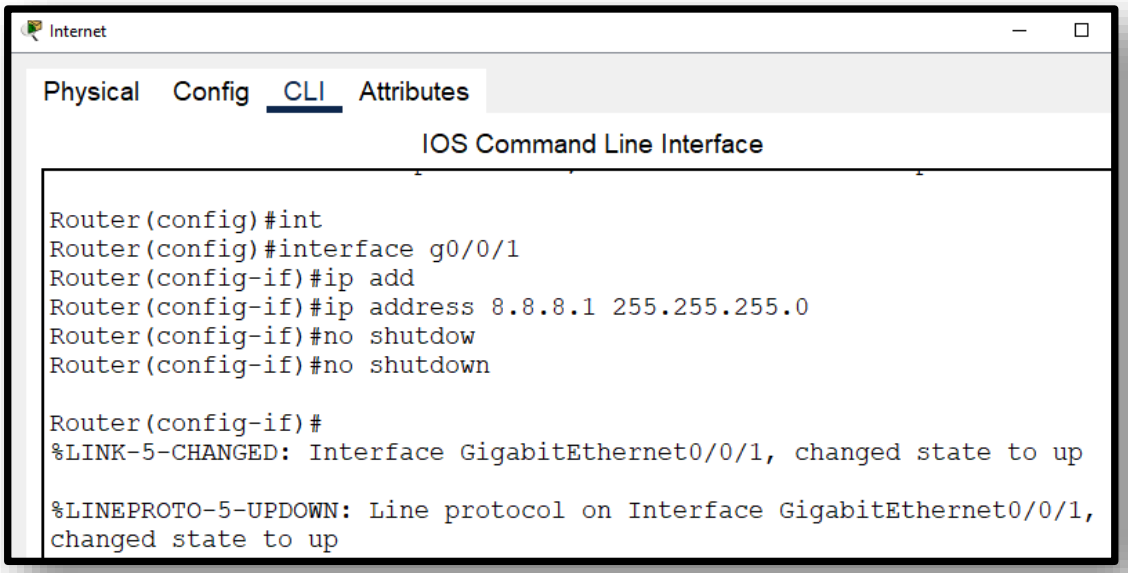


Step 2: Router Configuration IP and Subnet Mask

Click on Router ISR 4321 "Internet," CLI-tab, and follow the configuration below:(See Fig N° 2)

```
ena
configure terminal
interface g0/0/1
ip address 8.8.8.1 255.255.255.0
no shutdown
interface g0/0/0
ip address 209.165.200.225 255.255.255.252
no shutdown
```

Fig N° 2

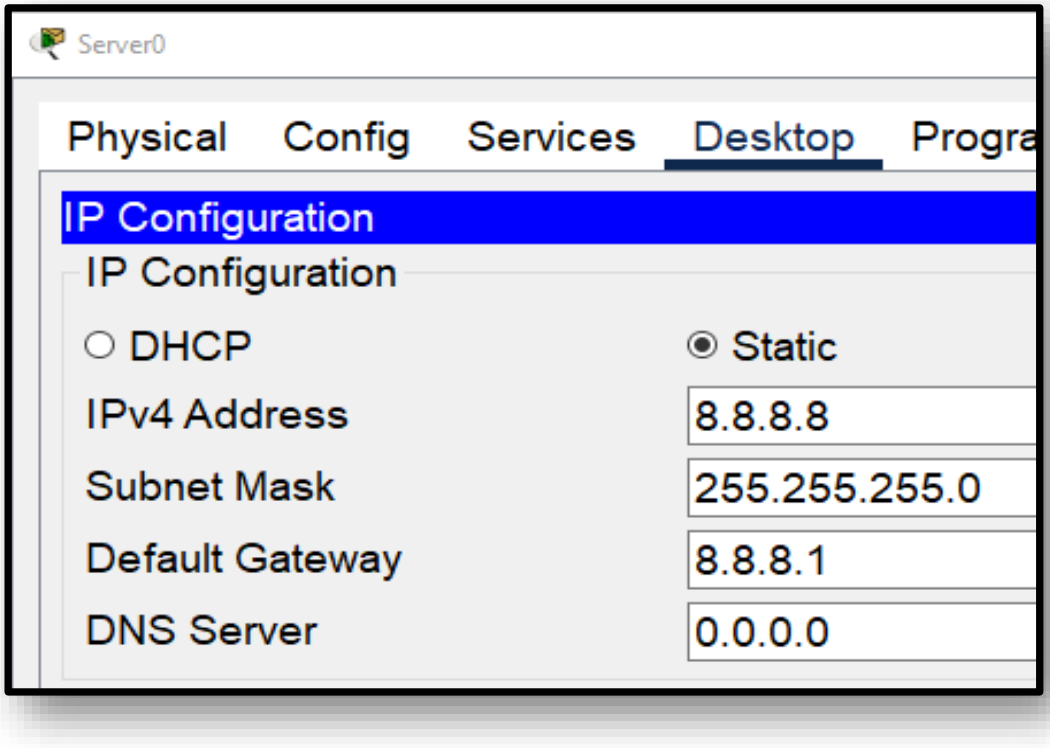

Internet
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router(config)#int
Router(config)#interface g0/0/1
Router(config-if)#ip add
Router(config-if)#ip address 8.8.8.1 255.255.255.0
Router(config-if)#no shutdow
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
Router(config)#interface g0/0/0
Router(config-if)#ip address 209.165.200.225 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up
Router(config-if)#

Step 3: Server0 Configuration IP, Subnet Mask, and Default Gateway

Click on Server0,Desktop-tab, IP Configuration, choose the radio button Static, and follow the configuration below:(See Fig N° 3)

IPv4 Address :8.8.8.0
Subnet Mask :255.255.255.0
Default Gateway :8.8.8.1

Fig N° 3



Step 4: 5506-X ASA0 Configuration
Click on ASA0, CLI-tab, and follow the configuration below:

4.1 Set up hostname and password, verify current configuration
(See Fig N° 4A, 4B and 4C)

```
enable
conf terminal
hostname ASA5506
enable password rowan123
show running
```

Fig N° 4A

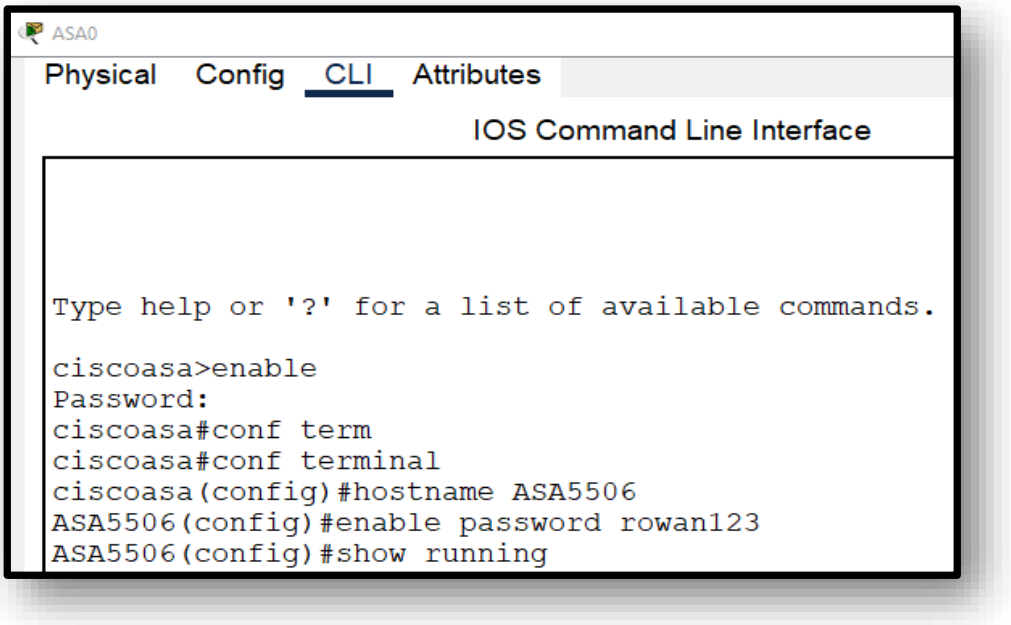


Fig N° 4B

```
ASA5506(config)#show running
: Saved
:
ASA Version 9.6(1)
!
hostname ASA5506
enable password QS5H9JV4DFwT0Jhy encrypted
names
!
interface GigabitEthernet1/1
 no nameif
 no security-level
 no ip address
 shutdown
!
interface GigabitEthernet1/2
 no nameif
 no security-level
 no ip address
 shutdown
!
interface GigabitEthernet1/3
 no nameif
 no security-level
 no ip address
 shutdown
!
```

```
interface GigabitEthernet1/4
 no nameif
 no security-level
 no ip address
 shutdown
!
interface GigabitEthernet1/5
 no nameif
 no security-level
 no ip address
 shutdown
!
interface GigabitEthernet1/6
 no nameif
 no security-level
 no ip address
 shutdown
!
interface GigabitEthernet1/7
 no nameif
 no security-level
 no ip address
 shutdown
!
interface GigabitEthernet1/8
 no nameif
 no security-level
 no ip address
 shutdown
!
```

Fig N° 4C

```
interface Management1/1
 management-only
 no nameif
 no security-level
 no ip address
 shutdown
!
!
!
!
!
!
!
class-map inspection_default
 match default-inspection-traffic
!
policy-map type inspect dns preset_dns_map
 parameters
  message-length maximum 512
policy-map global_policy
 class inspection_default
  inspect dns preset_dns_map
  inspect ftp
  inspect tftp
!
service-policy global_policy global
!
telnet timeout 5
ssh timeout 5
!
```

4.2 Set up interface GigabitEthernet1/2 (g1/2) and GigabitEthernet1/1 (g1/1).

GigabitEthernet1/2 . (See Fig N° 5A)

```
interface g1/2
nameif outside
security-level 0
ip address 209.165.200.226 225.255.255.252
no shutdown
```

Fig N° 5A

```
ASA5506(config)#int
ASA5506(config)#interface g1/2
ASA5506(config-if)#nameif outside
INFO: Security level for "outside" set to 0 by default.
ASA5506(config-if)#security-
ASA5506(config-if)#security-level 0
ASA5506(config-if)#ip address 209.165.200.226 255.255.255.252
ASA5506(config-if)#no shutdown
```

GigabitEthernet1/1.(See Fig N° 5B)

```
interface g1/1
nameif inside
security-level 100
ip address 192.168.1.1 225.255.255.0
no shutdown
exit
```

Fig N° 5B

```
ASA5506(config-if)#interface g1/1
ASA5506(config-if)#nameif inside
INFO: Security level for "inside" set to 100 by default.
ASA5506(config-if)#security-level 100
ASA5506(config-if)#ip address 192.168.1.1 255.255.255.0
ASA5506(config-if)#security-level 100
ASA5506(config-if)#no shutdown

ASA5506(config-if)#exit
ASA5506(config)#
```

4.3 Verify the new configuration.(See Fig N° 6A and 6B)

```
show interface ip brief
```

Fig N° 6A

```
ASA5506(config)#show interface ip brief
Interface              IP-Address      OK? Method Status                Protocol
Virtual0               127.1.0.1       YES unset  up                    up
GigabitEthernet1/1     192.168.1.1     YES manual up                    up
GigabitEthernet1/2     209.165.200.226 YES manual up                    up
GigabitEthernet1/3     unassigned      YES unset  administratively down down
GigabitEthernet1/4     unassigned      YES unset  administratively down down
GigabitEthernet1/5     unassigned      YES unset  administratively down down
GigabitEthernet1/6     unassigned      YES unset  administratively down down
GigabitEthernet1/7     unassigned      YES unset  administratively down down
GigabitEthernet1/8     unassigned      YES unset  administratively down down
Management1/1          unassigned      YES unset  administratively down down
Internal-Control1/1    127.0.1.1       YES unset  up                    up
Internal-Data1/1       unassigned      YES unset  up                    up
Internal-Data1/2       unassigned      YES unset  up                    up
Internal-Data1/3       unassigned      YES unset  up                    up
ASA5506(config)#
```

```
show ip address
```

Fig N° 6B

```
ASA5506(config)#show ip address
System IP Addresses:
Interface              Name                IP address      Subnet mask     Method
GigabitEthernet1/1     inside              192.168.1.1     255.255.255.0   manual
GigabitEthernet1/2     outside             209.165.200.226 255.255.255.252 manual
GigabitEthernet1/3                         unassigned      unassigned      unset
GigabitEthernet1/4                         unassigned      unassigned      unset
GigabitEthernet1/5                         unassigned      unassigned      unset
GigabitEthernet1/6                         unassigned      unassigned      unset
GigabitEthernet1/7                         unassigned      unassigned      unset
GigabitEthernet1/8                         unassigned      unassigned      unset
Management1/1                              unassigned      unassigned      unset

Current IP Addresses:
Interface              Name                IP address      Subnet mask     Method
GigabitEthernet1/1     inside              192.168.1.1     255.255.255.0   manual
GigabitEthernet1/2     outside             209.165.200.226 255.255.255.252 manual
GigabitEthernet1/3                         unassigned      unassigned      unset
GigabitEthernet1/4                         unassigned      unassigned      unset
GigabitEthernet1/5                         unassigned      unassigned      unset
GigabitEthernet1/6                         unassigned      unassigned      unset
GigabitEthernet1/7                         unassigned      unassigned      unset
GigabitEthernet1/8                         unassigned      unassigned      unset
Management1/1                              unassigned      unassigned      unset

ASA5506(config)#
```

4.4 Set up the dhcp configuration.(See Fig N° 7)

```
dhcp address 192.168.1.10-192.168.1.100 inside
dhcp dns 8.8.8.8
dhcp option 3 ip 192.168.1.1
dhcp enable inside
```

Fig N° 7

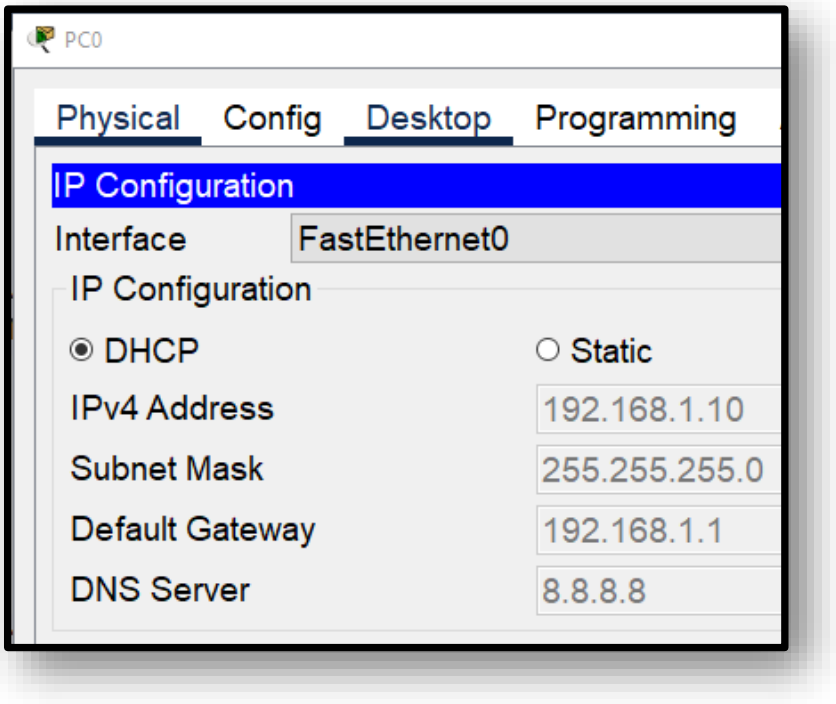
```
ASA5506(config)#dhcp address 192.168.1.10-192.168.1.100 inside



ASA5506(config)#dhcp dns 8.8.8.8
ASA5506(config)#dhcp op
ASA5506(config)#dhcp option 3 ip 192.168.1.1
ASA5506(config)#dhcp enable inside
ASA5506(config)#
```

Step 5: PC-PT PC0 Configuration
Click on PC0, Desktop-tab, and IP Configuration, and select the radio button DHCP. Verify the configuration below:(See Fig N° 8)

Fig N° 8



Step 6: 5506-X ASA0 Configuration II
Click on ASA0, CLI-tab, and follow the configuration below:

6.1 Route Configuration. (See Fig N° 9)

```
route outside 0.0.0.0 0.0.0.0 209.165.200.225
show route
```

Fig N° 9

```
ASA5506(config)#route outside 0.0.0.0 0.0.0.0 209.165.200.225
ASA5506(config)#show route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 209.165.200.225 to network 0.0.0.0

C    192.168.1.0 255.255.255.0 is directly connected, inside, GigabitEthernet1/1
     209.165.200.0/30 is subnetted, 2 subnets
C       209.165.200.0 255.255.255.252 is directly connected, outside,
GigabitEthernet1/2
C       209.165.200.224 255.255.255.252 is directly connected, outside,
GigabitEthernet1/2
S*   0.0.0.0/0 [1/0] via 209.165.200.225
ASA5506(config)#
```

6.2 Creation of an Object Network called INSIDE-NET and Network Address Translation (NAT). (See Fig N° 10)

```
object network INSIDE-NET
subnet 192.168.1.0 255.255.255.0
nat (inside,outside) dynamic interface
show nat
```

Fig N° 10

```
ASA5506(config)#object network INSIDE-NET
ASA5506(config-network-object)#subnet 192.168.1.0 255.255.255.0
ASA5506(config-network-object)#nat (inside,outside) dynamic interface
ASA5506(config-network-object)#show nat
Auto NAT Policies (Section 2)
1 (inside) to (outside) source dynamic INSIDE-NET interface
    translate_hits = 0, untranslate_hits = 0

ASA5506(config-network-object)#
```

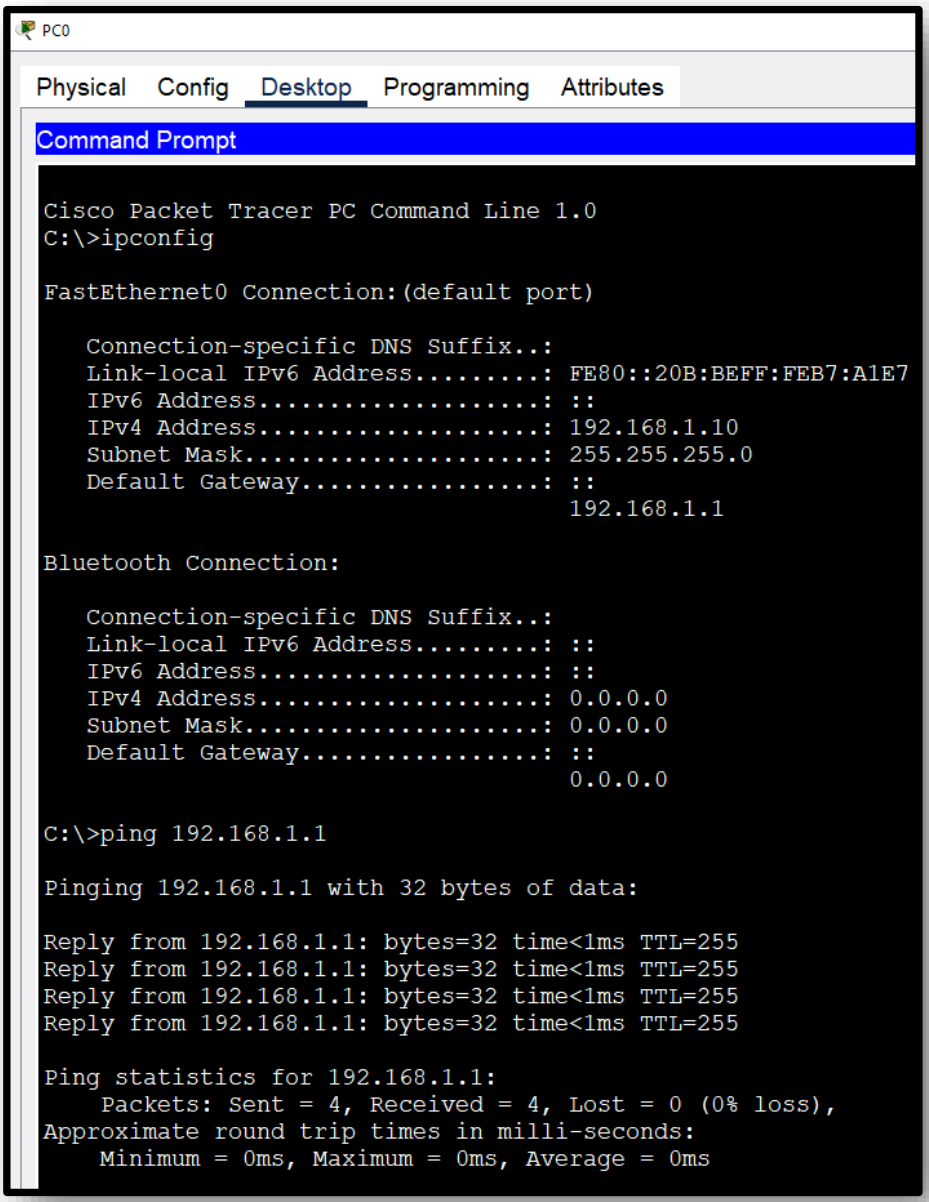
Step 7: Testing communication I

Click on PC0, Desktop-tab, Command Prompt, and follow the test communication below:

7.1 PC0 – Gig1/1. (See Fig N° 11A)

```
ipconfig
ping 192.168.1.1
```

Fig N° 11A



7.2 PC0 – Gig0/0/0. (See Fig N° 11B)

```
ping 209.165.200.225
```

Fig N° 11B

```
C:\>ping 209.165.200.225

Pinging 209.165.200.225 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 209.165.200.225:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

7.2 PC0 – Server0. (See Fig N° 11C)

```
ping 8.8.8.8
```

Fig N° 11C

```
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Step 8: 5506-X ASA0 Configuration III
Click on ASA0, CLI-tab, and follow the configuration below:

8.1 Checking current configuration of ASA0. (See Fig N° 12A, 12B)

```
exit
show running
```

Fig N° 12A

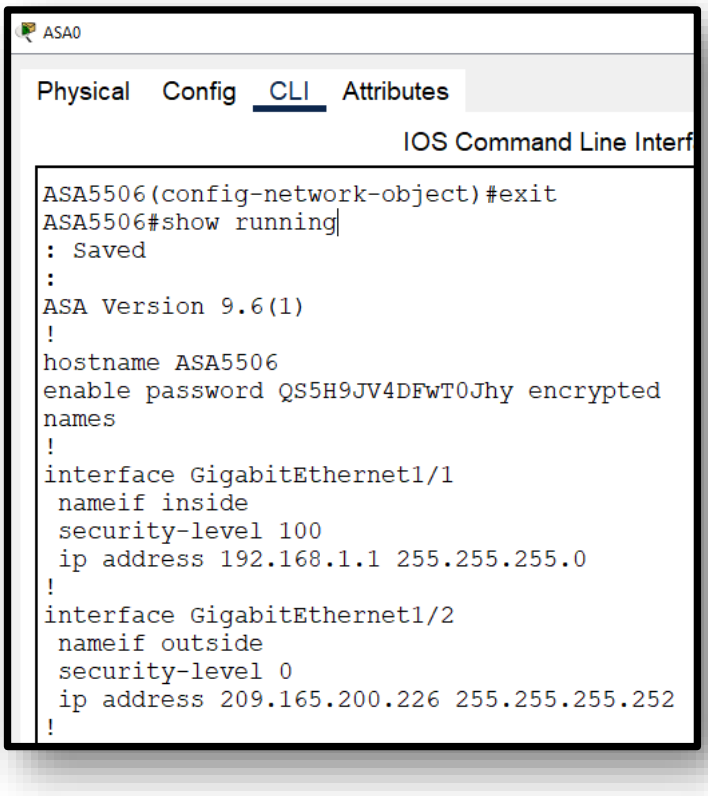
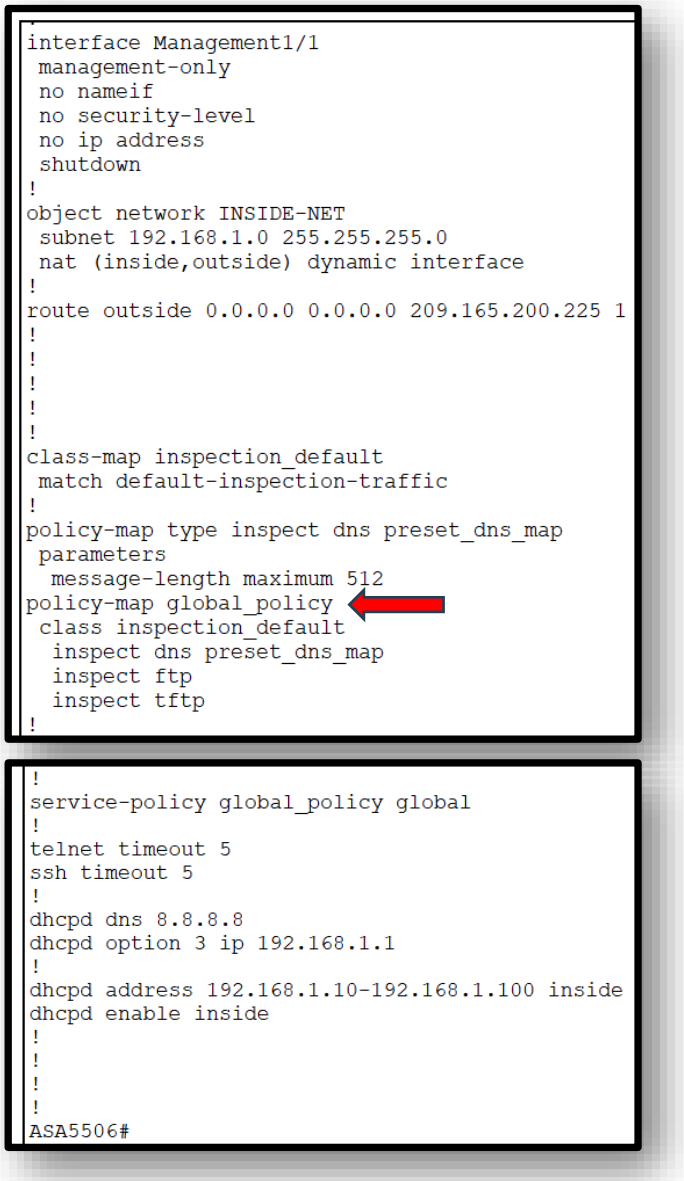


Fig N° 12B



```
interface Management1/1
 management-only
 no nameif
 no security-level
 no ip address
 shutdown
!
object network INSIDE-NET
 subnet 192.168.1.0 255.255.255.0
 nat (inside,outside) dynamic interface
!
route outside 0.0.0.0 0.0.0.0 209.165.200.225 1
!
!
!
!
!
class-map inspection_default
 match default-inspection-traffic
!
policy-map type inspect dns preset_dns_map
 parameters
  message-length maximum 512
policy-map global_policy
 class inspection_default
  inspect dns preset_dns_map
  inspect ftp
  inspect tftp
!
!
service-policy global_policy global
!
telnet timeout 5
ssh timeout 5
!
dhcpd dns 8.8.8.8
dhcpd option 3 ip 192.168.1.1
!
dhcpd address 192.168.1.10-192.168.1.100 inside
dhcpd enable inside
!
!
!
!
ASA5506#
```

8.2 Remove Policy-map global policy (See Fig N° 13A, and 13B)

```
confi terminal
no policy-map global_policy
show running
```

Fig N° 13A

```
ASA5506#conf terminal
ASA5506(config)#no pol
ASA5506(config)#no policy-map global_policy
```

```
ASA5506(config)#show running
: Saved
:
ASA Version 9.6(1)
!
hostname ASA5506
enable password QS5H9JV4DFwT0Jhy encrypted
names
!
interface GigabitEthernet1/1
 nameif inside
 security-level 100
 ip address 192.168.1.1 255.255.255.0
!
interface GigabitEthernet1/2
 nameif outside
 security-level 0
 ip address 209.165.200.226 255.255.255.252
!
```

Fig N° 13B

```
interface Management1/1
 management-only
 no nameif
 no security-level
 no ip address
 shutdown
!
object network INSIDE-NET
 subnet 192.168.1.0 255.255.255.0
 nat (inside,outside) dynamic interface
!
route outside 0.0.0.0 0.0.0.0 209.165.200.225 1
!
!
!
!
!
class-map inspection_default
 match default-inspection-traffic
!
policy-map type inspect dns preset_dns_map
 parameters
  message-length maximum 512
!
service-policy global_policy global
!
telnet timeout 5
ssh timeout 5
!
dhcpd dns 8.8.8.8
dhcpd option 3 ip 192.168.1.1
!
dhcpd address 192.168.1.10-192.168.1.100 inside
dhcpd enable inside
!
!
!
!
ASA5506(config)#
```

Note: To compare The Fig Nº 12B with Fig Nº 13B

8.2 Re-create Policy-map global policy, remove, and re-create no service-policy global_policy global (See Fig N° 14)

```
policy-map global policy
class inspection_default
inspect icmp
exit
no service-policy global_policy global
service-policy global_policy global
show running
```

Fig N° 14

```
ASA5506(config)#policy-map global_policy
ASA5506(config-pmap)#class inspection_default
ASA5506(config-pmap-c)#inspect icmp
ASA5506(config-pmap-c)#exit
ASA5506(config)#no service-policy global_policy global
ASA5506(config)#service-policy global_policy global
ASA5506(config)#
```

```
class-map inspection_default
 match default-inspection-traffic
!
policy-map type inspect dns preset_dns_map
 parameters
  message-length maximum 512
policy-map global_policy
 class inspection_default
  inspect icmp
!
service-policy global_policy global
!
telnet timeout 5
ssh timeout 5
!
dhcpd dns 8.8.8.8
dhcpd option 3 ip 192.168.1.1
!
dhcpd address 192.168.1.10-192.168.1.100 inside
dhcpd enable inside
!
!
!
!
ASA5506(config)#
```

Step 9: Testing communication II
Click on PC0, Desktop-tab, Command Prompt, and follow the test communication below:

9.1 PC0 – Gig0/0/0. (See Fig N° 15A)

Fig N° 15A

```
C:\>ping 209.165.200.225

Pinging 209.165.200.225 with 32 bytes of data:

Reply from 209.165.200.225: bytes=32 time<1ms TTL=254
Reply from 209.165.200.225: bytes=32 time<1ms TTL=254
Reply from 209.165.200.225: bytes=32 time<1ms TTL=254
Reply from 209.165.200.225: bytes=32 time<1ms TTL=254

Ping statistics for 209.165.200.225:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

9.2 PC0 – Serve0. (See Fig N° 15B)

Fig N° 15B

```
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

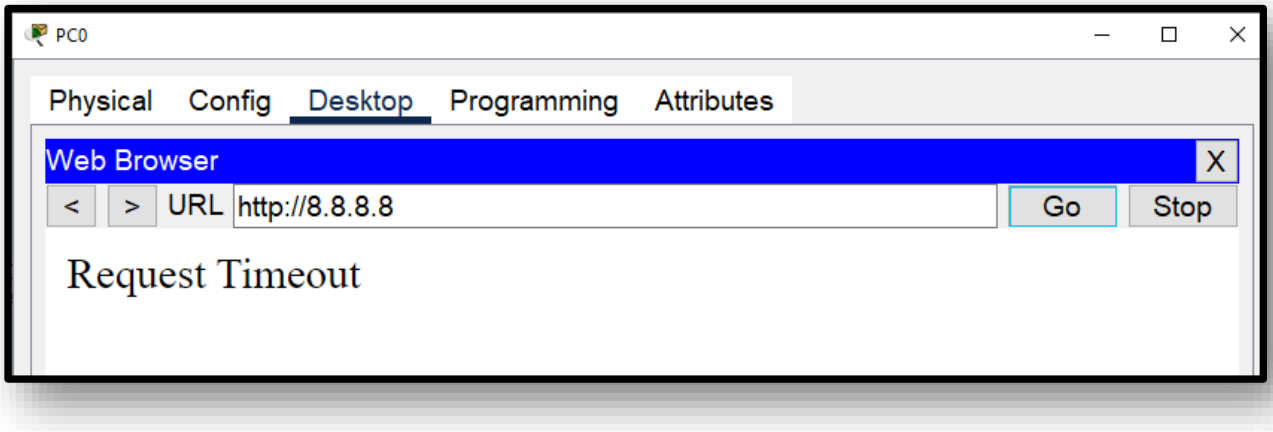
Reply from 8.8.8.8: bytes=32 time<1ms TTL=126
Reply from 8.8.8.8: bytes=32 time<1ms TTL=126
Reply from 8.8.8.8: bytes=32 time<1ms TTL=126
Reply from 8.8.8.8: bytes=32 time<1ms TTL=126

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Step 10: Testing communication III
Click on PC0, Desktop-tab, Web Browser, and follow the test communication below:

10.1 PC0 – Server0 (Service HTTP). (See Fig N° 16)

Fig N° 16



Step 11: 5506-X ASA0 Configuration IV
Click on ASA0, CLI-tab, and follow the configuration below:

11.1 Activate inspect http. (See Fig N° 17)

```
policy-map global policy
class inspection_default
inspect http
exit
show running
```

Fig N° 17

```
ASA5506(config)#policy-map global_policy
ASA5506(config-pmap)#class inspection_default
ASA5506(config-pmap-c)#inspect http
ASA5506(config-pmap-c)#exit
ASA5506(config)#show running
```

```
class-map inspection_default
 match default-inspection-traffic
!
policy-map type inspect dns preset_dns_map
 parameters
  message-length maximum 512
policy-map global_policy
 class inspection_default
  inspect http
  inspect icmp
!
service-policy global_policy global
!
telnet timeout 5
ssh timeout 5
!
dhcpd dns 8.8.8.8
dhcpd option 3 ip 192.168.1.1
!
dhcpd address 192.168.1.10-192.168.1.100 inside
dhcpd enable inside
!
!
!
!
ASA5506(config)#
```

Step 12: Testing communication IV

Click on PC0, Desktop-tab, Web Browser, and follow the test communication below:

12.1 PC0 – Server0 (Service HTTP). (See Fig N° 18)

Fig N° 18

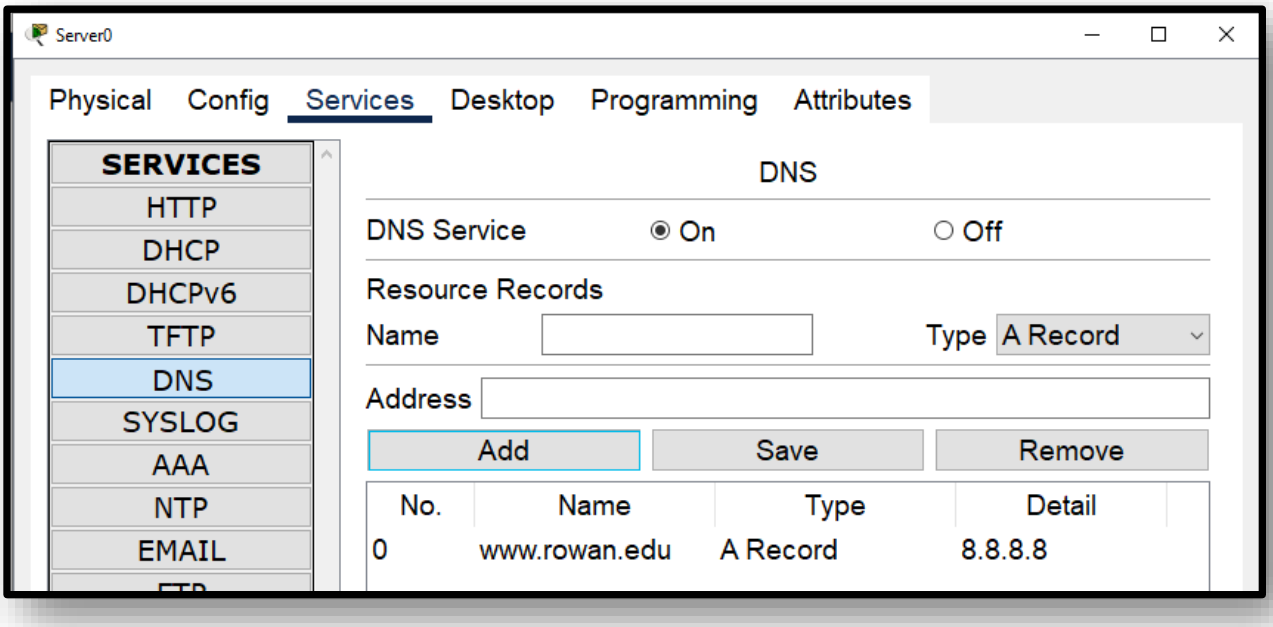


Step 13: Set up DNS Service at Server0

Click on Server0, Service-tab, Service-Module/DNS, and follow the configuration below: (See Fig N° 19)

DNS Service : ON
Resource Records/Name : www.rowan.edu
Address : 8.8.8.8
Click on Add Button

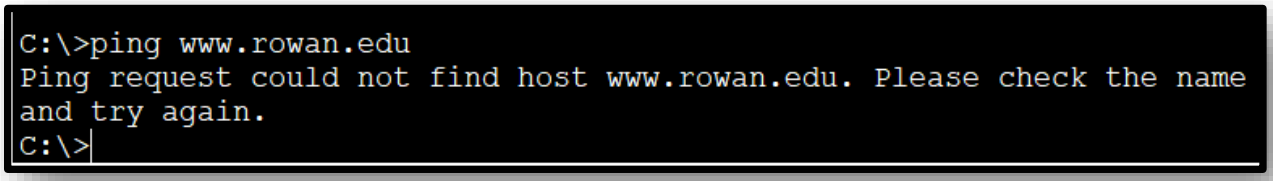
Fig N° 19



Step 14: Testing communication V
Click on PC0, Desktop-tab, Command Prompt, and follow the test communication below:

14.1 PC0 – Server0 (DNS www.rowan.edu). (See Fig N° 20)

Fig N° 20



Step 15: 5506-X ASA0 Configuration V
Click on ASA0, CLI-tab, and follow the configuration below:

15.1 Activate inspect dns. (See Fig N° 21)

```
policy-map global_policy
class inspection_default
inspect dns preset_dns_map
exit
show running
```

Fig N° 21

```
ASA5506(config)#policy-map global_policy
ASA5506(config-pmap)#class inspection_default
ASA5506(config-pmap-c)#inspect dns preset_dns_map
ASA5506(config-pmap-c)#exit
ASA5506(config)#show running
```

```
class-map inspection_default
 match default-inspection-traffic
!
policy-map type inspect dns preset_dns_map
 parameters
  message-length maximum 512
policy-map global_policy
 class inspection_default
  inspect dns preset_dns_map
  inspect http
  inspect icmp
!
service-policy global_policy global
!
telnet timeout 5
ssh timeout 5
!
dhcpd dns 8.8.8.8
dhcpd option 3 ip 192.168.1.1
!
dhcpd address 192.168.1.10-192.168.1.100 inside
dhcpd enable inside
!
!
!
!
ASA5506(config)#
```

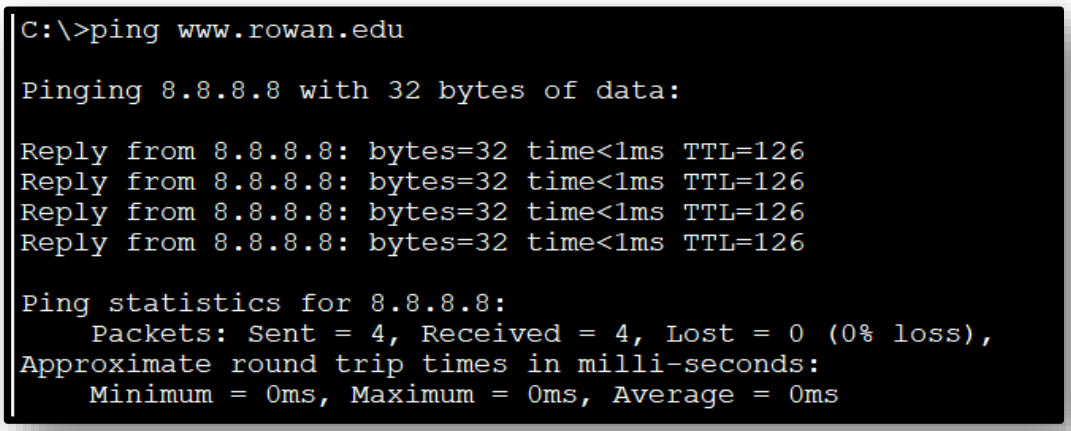
Step 16: Testing communication VI

Click on PC0, Desktop-tab, Command Prompt or Web Browser, and follow the test communication below:

16.1 PC0 – Server0 (DNS www.rowan.edu) by command prompt.

(See Fig N° 22A)

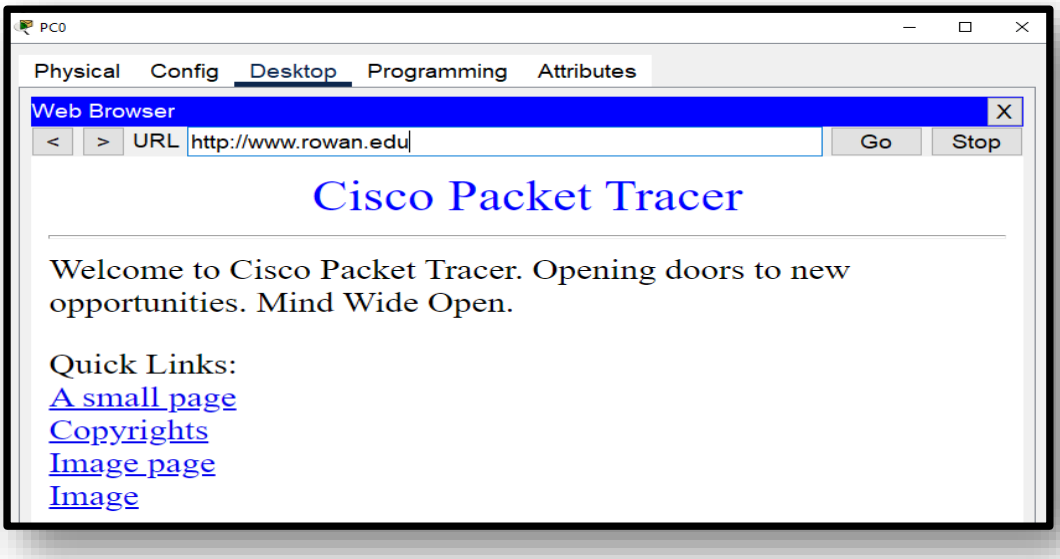
Fig N° 22A



16.2 PC0 – Server0 (DNS www.rowan.edu) by Web Browser.

(See Fig N° 22B)

Fig N° 22B



Step 17: 5506-X ASA0 Configuration VI

Click on ASA0, CLI-tab, and follow the configuration below:

17.1 Activate inspect SSH. (See Fig N° 23)

```
domain-name cisco.com
username admin password cisco123!
aaa authentication ssh console local
crypto key generate rsa modulus 1024
yes
ssh 192.168.1.0 255.255.255.0 inside
ssh timeout 10
```

Fig N° 23

```
ASA5506(config)#domain-name cisco.com
ASA5506(config)#username admin password cisco123!
ASA5506(config)#aaa authentication ssh console local
ASA5506(config)#crypto key generate rsa modulus 1024
WARNING: You have a RSA keypair already defined named <Default-RSA-Key>.

Do you really want to replace them? [yes/no]: yes
Keypair generation process begin. Please wait...


ASA5506(config)#ssh 192.168.1.0 255.255.255.0 inside
ASA5506(config)#ssh timeout ?

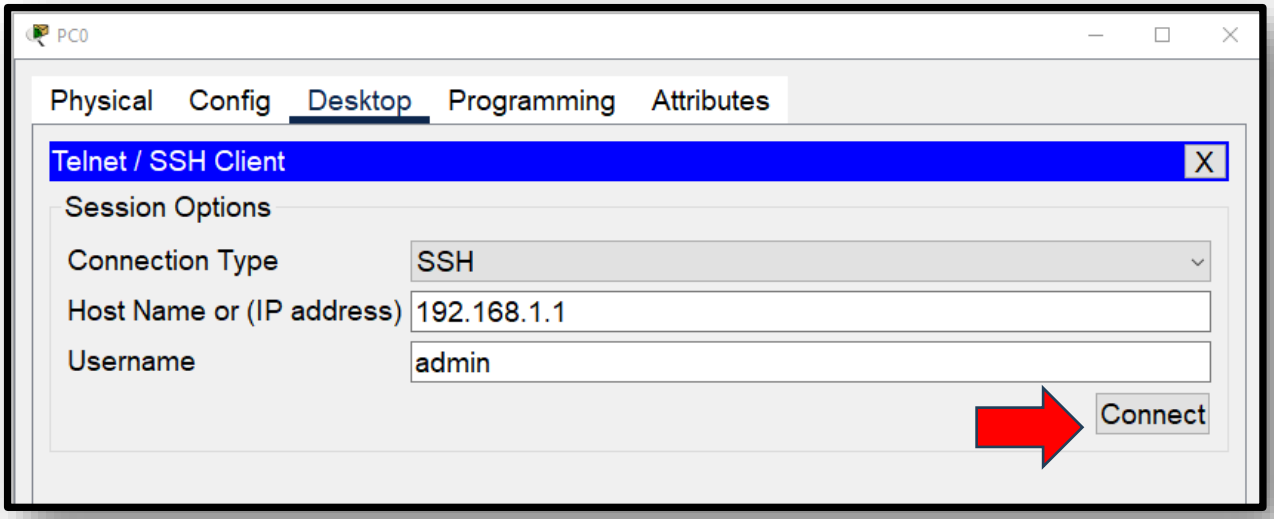
configure mode commands/options:
  <1-60>  Idle time in minutes after which a ssh session will be closed
  <cr>
ASA5506(config)#ssh timeout 10
ASA5506(config)#
```

Step 18: Testing communication VII

Click on PC0, Desktop-tab, Telnet/SSH Client, and follow the test communication below:

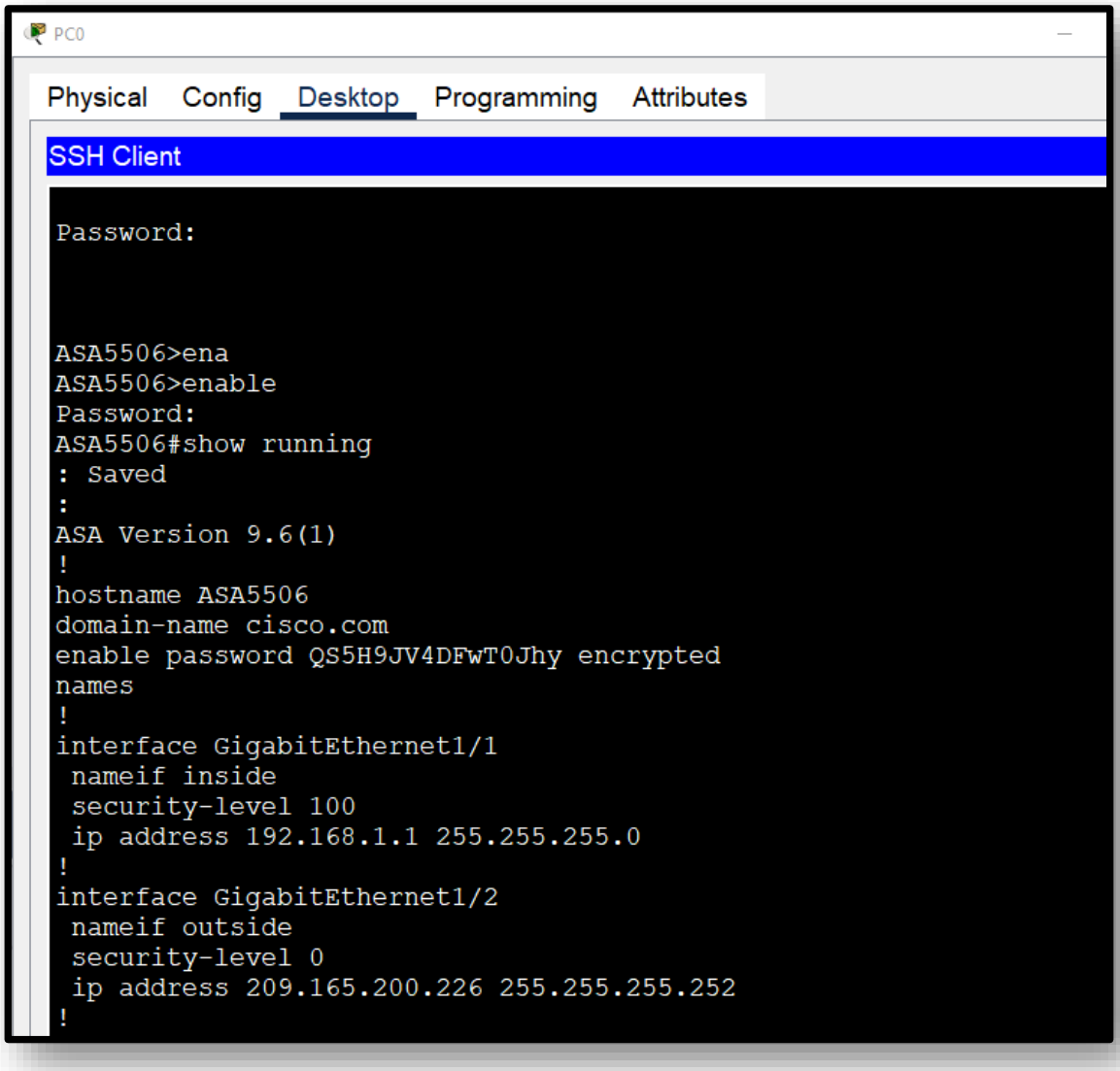
18.1 PC0 – ASA0 (See Fig N° 24A and 24B)

Fig N° 24




```
Password:cisco123!
enable
password:rowan123
show running
```

Fig N° 24B


PC0
Physical
Config
Desktop
Programming
Attributes
SSH Client
Password:
ASA5506>ena
ASA5506>enable
Password:
ASA5506#show running
: Saved
:
ASA Version 9.6(1)
!
hostname ASA5506
domain-name cisco.com
enable password QS5H9JV4DFwT0Jhy encrypted
names
!
interface GigabitEthernet1/1
 nameif inside
 security-level 100
 ip address 192.168.1.1 255.255.255.0
!
interface GigabitEthernet1/2
 nameif outside
 security-level 0
 ip address 209.165.200.226 255.255.255.252
!

* * *